Example: Particle Filter
Reference: Thrun, Burgard, Fox, "Probabilistic Robotics," MIT Press 2006

1: **Algorithm MCL**($\mathcal{X}_{t-1}, u_t, z_t, m$)**:**
2: $\bar{\mathcal{X}}_t = \mathcal{X}_t = \emptyset$
3: for $m = 1$ to M do
4: $x_t^{[m]} = \textbf{sample_motion_model}(u_t, x_{t-1}^{[m]})$
5: $w_t^{[m]} = \textbf{measurement_model}(z_t, x_t^{[m]}, m)$
6: $\bar{\mathcal{X}}_t = \bar{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$
7: endfor
8: for $m = 1$ to M do
9: draw i with probability $\propto w_t^{[i]}$
10: add $x_t^{[i]}$ to $\mathcal{X}_t$
11: endfor
12: return $\mathcal{X}_t$

Table 8.2 MCL, or Monte Carlo Localization, a localization algorithm based on particle fi lters.

better the results. Fine-grained approximations also tend to suffer less from "catastrophic" failures where the robot's belief differs significantly from its actual position.

8.3 MONTE CARLO LOCALIZATION

8.3.1 The MCL Algorithm

We will now turn our attention to a popular localization algorithm which represents the belief $bel(x_t)$ by particles. The algorithm is called *Monte Carlo Localization*, or *MCL*. Like grid-based Markov localization, MCL is applicable to both local and global localization problems. Despite its relatively short existence, MCL has already become one of the most popular localization algorithms in robotics. It is easy to implement, and tends to work well across a broad range of localization problems.

Table 8.2 shows the basic MCL algorithm, which is obtained by substituting the appropriate probabilistic motion and perceptual models into the algorithm **particle_filters** (Table 4.3 on page 78). The basic MCL algorithm represents the belief $bel(x_t)$ by a set of M particles $\mathcal{X}_t = \{x_t^{[1]}, x_t^{[2]}, \ldots, x_t^{[M]}\}$. Lines 4 in our algorithm (Table 8.2) sam-

ples from the motion model, using particles from present belief as starting points. The beam measurement model is then applied in line 5 to determine the importance weight of that particle. The initial belief $bel(x_0)$ is obtained by randomly generating M such particles from the prior distribution $p(x_0)$, and assigning the uniform importance factor M^{-1} to each particle. As in grid localization, the functions **motion_model**, and **measurement_model**, may be implemented by any of the motion models in Chapter 5, and measurement models in Chapter 6, respectively.

Figure 8.11 illustrates MCL using the one-dimensional hallway example. The initial global uncertainty is achieved through a set of pose particles drawn at random and uniformly over the entire pose space, as shown in Figure 8.11a. As the robot senses the door, line 5 of in the algorithm **MCL** assigns importance factors to each particle. The resulting particle set is shown in Figure 8.11b. The height of each particle in this figure shows its importance weight. It is important to notice that this set of particles is identical to the one in Figure 8.11a—the only thing modified by the measurement update are the importance weights.

Figure 8.11c shows the particle set after resampling (line 8-11 in the algorithm **MCL**) and after incorporating the robot motion (line 4). This leads to a new particle set with uniform importance weights, but with an increased number of particles near the three likely places. The new measurement assigns non-uniform importance weights to the particle set, as shown in Figure 8.11d. At this point, most of the cumulative probability mass is centered on the second door, which is also the most likely location. Further motion leads to another resampling step, and a step in which a new particle set is generated according to the motion model (Figure 8.11e). As should be obvious from this example, the particle sets approximate the correct posterior, as would be calculated by an exact Bayes filter.

Figure 8.12 shows the result of applying MCL in an actual office environment, for a robot equipped with an array of sonar range finders. The figure depicts particle sets after 5, 28, and 55, meters of robot motion, respectively. Each particle set is defined over the 3-dimensional pose space, although only the x- and y-coordinates of each particle are shown. A second illustration is provided in Figure 8.13, here using a camera pointed towards the ceiling, and a measurement model that relates the brightness in the center of the image to a previously acquired ceiling map.

8.3.2 Properties of MCL

MCL can approximate almost any distribution of practical importance. It is not bound to a limited parametric subset of distributions, as was the case for EKF localization.

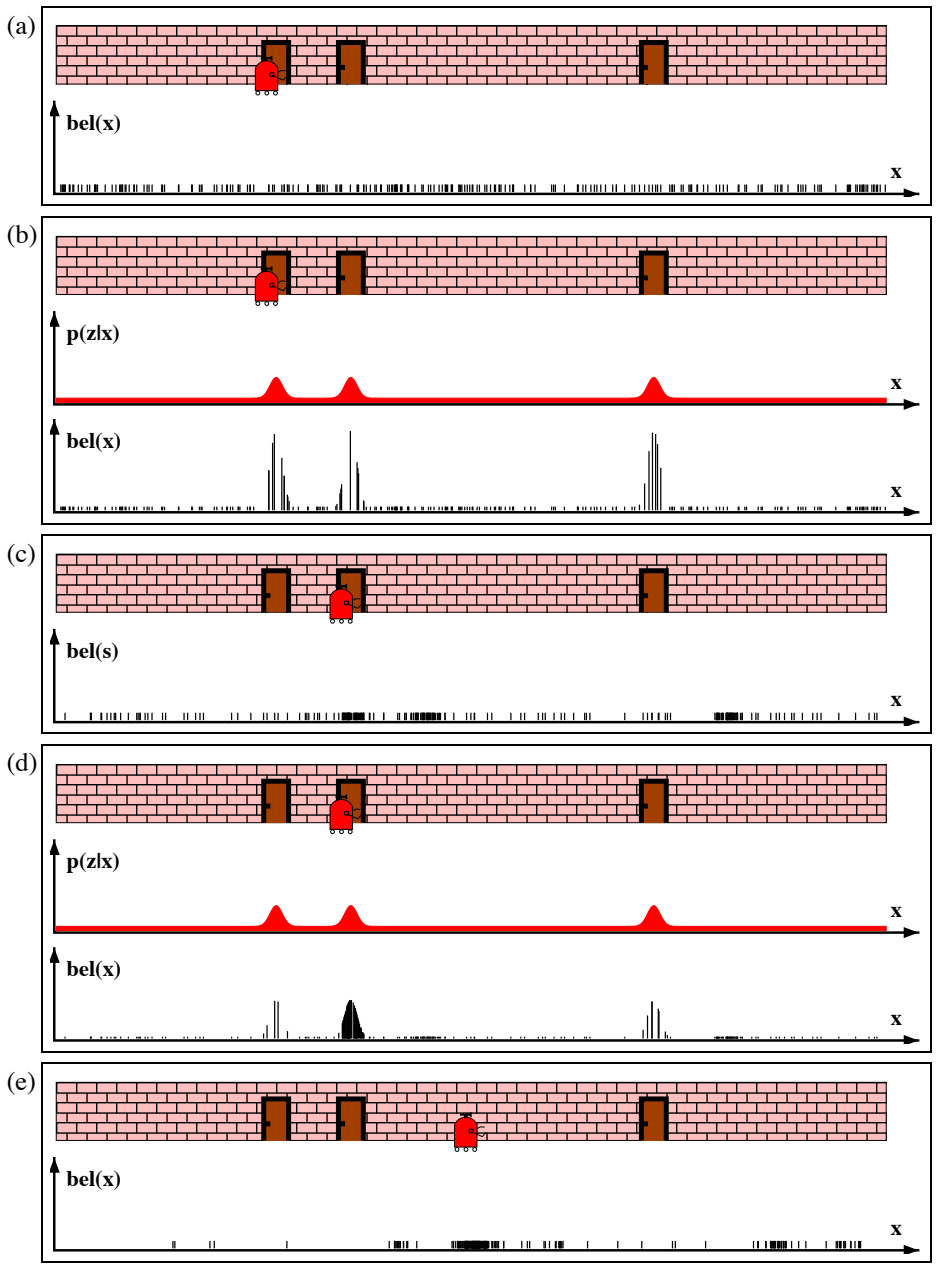


Figure 8.11 Monte Carlo Localization, a particle filter applied to mobile robot localization.

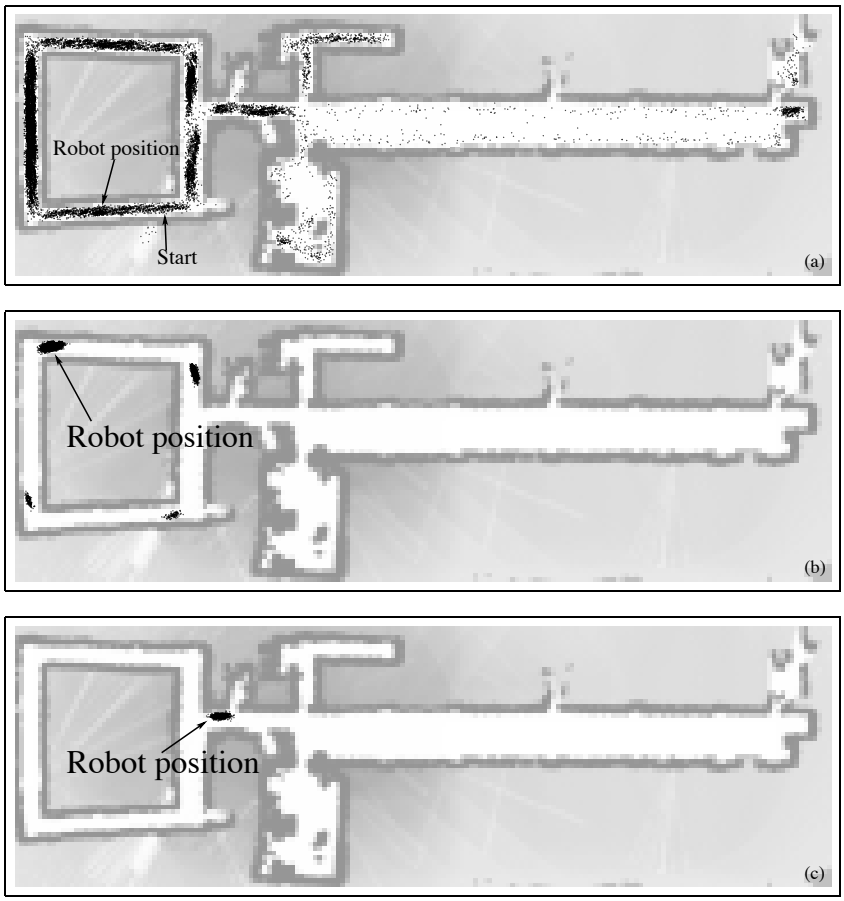


Figure 8.12 Illustration of Monte Carlo localization: Shown here is a robot operating in an office environment of size 54m $\times$ 18m. (a) After moving 5m, the robot is still highly uncertain about its position and the particles are spread trough major parts of the free-space. (b) Even as the robot reaches the upper left corner of the map, its belief is still concentrated around four possible locations. (c) Finally, after moving approximately 55m, the ambiguity is resolved and the robot knows where it is. All computation is carried out in real-time on a low-end PC.

Example: Particle Filter
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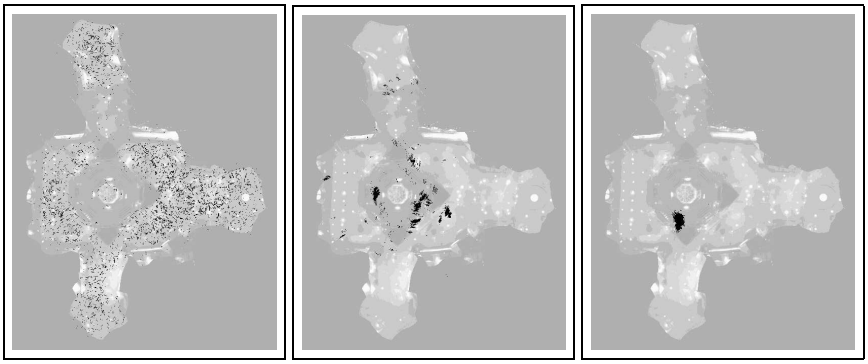


Figure 8.13 Global localization using a camera pointed at the ceiling.

The accuracy of the approximation is easily determined by the size of the particle set M. Increasing the total number of particles increases the accuracy of the approximation. The number of particles M is a parameter that enables the user to trade off the accuracy of the computation and the computational resources necessary to run MCL. A common strategy for setting M is to keep sampling until the next pair u_t and z_t has arrived. In this way, the implementation is adaptive with regards to the computational resources: the faster the underlying processor, the better the localization algorithm. Such a resource-adaptivity is difficult to achieve for grid localization and Gaussian techniques.

A final advantage of MCL pertains to the non-parametric nature of the approximation. As our illustrative results suggest, MCL can represent complex multi-modal probability distributions, and blend them seamlessly with focused Gaussian-style distributions. This provides MCL with the ability to solve global localization problems with high-accuracy position tracking.

8.3.3 Random Particle MCL: Recovery from Failures

MCL, in its present form, solves the global localization problem but cannot recover from robot kidnapping, or global localization failures. This quite obvious from the results in Figure 8.12: As the position is acquired, particles at places other than the